Cam Morreale

Reverse-engineers broken data systems into scalable decision infrastructure

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EDUCATION

Worcester Polytechnic Institute — MS Data Science (May 2026), BS Data Science (May 2025)

SKILLS

Languages & Development — Python, Pandas, NumPy, SQL, R, Power BI, React, TypeScript, Git

Data Engineering & Systems — ETL/ELT Pipelines, Data Modeling, Schema Design, Data Ingestion & Normalization, Async I/O (aiohttp)

Machine Learning & LLM Systems — Transformer Models (BERT, RoBERTa), Named Entity Recognition (NER), Retrieval-Augmented Generation (RAG), Embeddings (SentenceTransformers), Semantic Search, Prompt Engineering, Model Evaluation (Precision/Recall/F1), Hallucination Detection, Hugging Face, spaCy

Cloud & Infrastructure — AWS (Lambda, API Gateway, RDS, S3, IAM), Oracle Database, Serverless Architectures, Docker, REST APIs, IPFS, Filecoin, Web3.Storage

PROFESSIONAL EXPERIENCE

Wellington Management — Data Architecture & Engineering Intern | Boston, MA Jun–Aug 2025

- Reconstructed the data architecture powering a Python-based investment decision platform supporting $153B in insurance assets across 23 countries, replacing a fragmented Excel workflow used by ~160 investment professionals and enabling **~$8M in annual operational efficiency gains**.
- Reverse-engineered an undocumented Oracle environment (~250 tables), mapping business logic to the data layer by prioritizing key tables to reconstruct schema and data lineage under incomplete visibility.
- Diagnosed and resolved a missing upstream dataset as the root cause of system-wide pipeline failures, restoring execution by re-establishing critical data dependencies with minimal downstream changes.
- Co-designed an analytics-override framework balancing model integrity with user flexibility, enabling controlled input adjustments with validation and auditability for real-world investment decision workflows.
- Developed a system-level architecture mapping data pipelines, transformation logic, and user interaction layers, enabling reliable integration and clarifying dependencies across the platform.

MassDEP — Technical Business Analyst (Data & AI Systems) | Worcester, MA Jan–May 2025

- Owned retrieval, extraction, and validation layers of an LLM pipeline, converting unstructured reports into Power BI–ready datasets and reducing manual reporting effort by ~40% across 50+ projects.
- Reduced hallucinated outputs in LLM-generated data by implementing a retrieval-based grounding system using embeddings and semantic search, improving reliability of extracted information for analytics.
- Optimized entity extraction accuracy across inconsistent report formats by building a hybrid NER pipeline with validation logic, reducing missing and incorrect data in structured outputs.
- Standardized model outputs into structured CSV datasets and integrated them into Power BI, enabling analysis of extracted data across documents.
- Architected page-level inspection and validation logic to detect unreadable outputs, surface failure states, and quantify system reliability (97% successful extraction rate).

Embue — Data Systems Architecture Lead | Worcester, MA Aug–Dec 2022

- Addressed lack of tamper-resistant storage in IoT systems by architecting a distributed data infrastructure, enabling verifiable data integrity and reliable ingestion across decentralized environments.
- Designed a layered system separating encryption, storage, verification, and access control, improving scalability and security across the data pipeline.
- Identified scalability limitations of on-chain storage and implemented a hybrid architecture using IPFS for off-chain storage and Filecoin for verification, balancing performance with integrity guarantees.
- Enabled secure ingestion and retrieval of IoT data across decentralized systems by building an encrypted pipeline, ensuring confidentiality while maintaining verifiable data integrity.
- Deployed and validated decentralized infrastructure (IPFS nodes, Filecoin client), confirming end-to-end integrity through content-addressed retrieval and controlled decryption.

PROJECTS & HACKATHONS

BoardGameGeek Platform — Data Engineering & System Development (WPI) Aug–Dec 2025

- Improved reliability and throughput of API-constrained data ingestion by engineering a 3-stage asynchronous ELT pipeline, ensuring consistent collection of external data across ~150 entities.
- Standardized and integrated heterogeneous datasets by architecting data transformation workflows, reducing inconsistencies and improving downstream data usability.
- Modeled recursive relationships between entities to improve data integrity and support more complex queries beyond flat relational structures.
- Integrated backend data pipelines with AWS Lambda services and a React-based frontend, enabling full-stack functionality from ingestion to user interaction.

Oasis — EasyA x Polkadot Hackathon (1st place) | System Design & Business Logic Lead Jun 2023

- Led system design for a decentralized platform in a 48-hour hackathon (**1st out of 100+ teams**), translating governance and incentive requirements into a clear, executable architecture under tight constraints.
- Optimized token-based incentive mechanisms and governance structures, aligning system behavior with user participation and platform sustainability.
- Translated abstract economic and system requirements into a functional architecture, prioritizing clarity, feasibility, and rapid execution.

OneWorld — EasyA x HackBoston (3rd place) | System Architecture & Incentive Design Lead Oct 2022

- Designed and delivered a blockchain-based carbon market in a competitive hackathon (**Top 3 out of 100+ teams**), prioritizing business viability and incentive-aligned system design under time constraints.
- Modeled emissions permits as tokenized financial assets with lifecycle constraints, aligning system behavior with real-world regulatory and market dynamics.
- Engineered supply and pricing mechanisms (deterministic issuance decay, tiered marginal pricing) to control emissions supply and align long-term incentives.
- Architected on-chain infrastructure enabling real-time auditability of emissions issuance, trading, and retirement, reducing information asymmetry in carbon markets.

SafePool — DeFi Infrastructure (Avalanche L1 Hackathon) | System Design Lead Mar 2025

- Co-designed a decentralized financial system to formalize informal credit markets (ROSCAs), addressing counterparty risk, lack of transparency, and enforcement limitations in ~$1.5T+ global capital flows.
- Reduced counterparty risk in pooled savings systems by architecting smart contract–based financial primitives, enabling transparent and enforceable contribution and payout mechanisms.
- Enabled scalable deployment of a decentralized credit system by building infrastructure on Avalanche L1, optimizing for transaction throughput, settlement finality, and interoperability in underbanked markets.

FINANCIAL SYSTEMS & MARKETS

Quantitative Market Modeling — Multi-Strategy Trading Simulation (WPI) Jan–May 2022

- Pioneered trading simulations beating S&P 500 (+24.73%, +9.20% vs. +1.36%) over a 7-week period.
- Built a position-trading portfolio using DCF valuation, price-to-book analysis, and macroeconomic sector screening to identify undervalued equities in energy and emerging sectors.
- Analyzed options pricing dynamics and volatility-driven distortions in risk-reward.

Quantitative Market Research — Cryptocurrency ML Models (WPI) Mar–May 2024

- Conducted a meta-analysis of machine learning approaches for cryptocurrency trading, evaluating models including reinforcement learning (AdaBoost-LSTM), XGBoost, and SVM across profitability and risk-adjusted return metrics.
- Designed feature engineering and evaluation frameworks using technical indicators (RSI, MACD, ROC) and performance metrics (Sharpe ratio, drawdown, predictive accuracy).